2024 SSMO Speed Round Solutions

SMO Team

SR 1: Find the sum of the distinct prime factors of $2024^2 - 1$.

Answer: $\boxed{32}$

Solution: Note that $2024^2 - 1$ can be factored as $(2024 - 1)(2024 + 1) = (2023)(2025)$. Then, 2023 can be factored as $7 \cdot 17^2$ and 2025 can be factored as $45^2 = 3^4 \cdot 5^2$. Thus,

$$2024^2 - 1 = 2023 \cdot 2025 = 3^4 \cdot 5^2 \cdot 7 \cdot 17^2,$$

meaning the sum of the distinct prime factors is $3 + 5 + 7 + 17 = \boxed{32}$.

SR 2: Gracie's students play with some toys. When 4 or 5 students are present, the toys can be equally distributed to everyone. However, when there are only 3 students, there is one toy left over after giving everyone the same number of toys. What is the least possible number of toys that Gracie could have?

Answer: $\boxed{40}$

Solution: Suppose that there are x toys. Since x toys can be split equally when 4 or 5 students are present, x must be a multiple of $\text{lcm}(4, 5) = 20$. So, x can be $20, 40, 60, \ldots$. Now, since there is one toy left over when the toys are split when 3 students are present, x must be one greater than a multiple of 3. As $20 = 3 \cdot 6 + 2$ and $40 = 3 \cdot 13 + 1$, we conclude that the answer is $\boxed{40}$.

SR 3: The polynomial $x^3 - 15x^2 + 4x + 4$ has distinct real roots r, s, and t. Find the value of

$$\left|(r^2 + s^2 + t^2)(rst)\right|.$$

Answer: $\boxed{868}$

Solution: From Vieta's formulas, we have $r + s + t = 15$, $rs + rt + st = 4$, and $rst = -4$. Now, note that

$$\begin{aligned}\left|\left(r^2 + s^2 + t^2\right)(rst)\right| &= \left|\Big((r + s + t)^2 - 2(rs + rt + st)\Big)(rst)\right| \\ &= \left|\left(15^2 - 2 \cdot 4\right)(-4)\right| \\ &= |217 \cdot (-4)| = \boxed{868}.\end{aligned}$$

SR 4: Sam wants to read the Harry Potter and Warriors books. There are 7 Harry Potter books that must be read in a specific order, and there are 6 Warriors books that also must

be read in a specific order; however, he can read the two series at the same time. For example, he could read the first three Harry Potter books, then the first five Warriors books, then the remaining Harry Potter books, and finally the last Warriors book. In how many different orders can Sam read the books?

Answer: $\boxed{1716}$

Solution: Note that there are 13! ways to order the books if each series could be read in any order. However, only one of the 7! orders of the 7 Harry Potter books is allowed, and only one of the 6! orders of the 6 Warriors books is allowed. So, the answer is

$$\frac{13!}{7!\,6!} = \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{720} = 13 \cdot 12 \cdot 11 = \boxed{1716}.$$

SR 5: Let $\triangle ABC$ and $\triangle ADC$ be right triangles, such that $\angle ABC = \angle ADC = 90°$. Given that $\angle ACB = 30°$ and $BC = 3\sqrt{3}$, find the maximum possible length of BD.

Answer: $\boxed{6}$

Solution: Since $\angle ABC = \angle ADC = 90°$, the points B and D both lie on the circle with diameter AC. Since $\angle ACB = 30°$ and $BC = 3\sqrt{3}$, we have

$$AC = \frac{BC}{\cos \angle ACB} = \frac{3\sqrt{3}}{\frac{\sqrt{3}}{2}} = 6.$$

So, $BD \leq 6$, as no chord of a circle is longer than its diameter. Equality holds when D is the point diametrically opposite B, which is neither A nor C. Thus, the maximum possible length of BD is $\boxed{6}$.

SR 6: There are 4 people and 4 houses. Each person independently and uniformly at random chooses a house to live in. The expected number of inhabited houses can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{239}$

Solution: For each house, the probability that it is uninhabited is $\left(\frac{3}{4}\right)^4$. So, the probability that it is inhabited is $1 - \left(\frac{3}{4}\right)^4 = \frac{175}{256}$. Thus, the expected number of inhabited houses is

$$4 \cdot \left(\frac{175}{256}\right) = \frac{175}{64} \implies 175 + 64 = \boxed{239}.$$

SR 7: Let S denote the set of all ellipses centered at the origin with axes AB and CD, where $A = (-x, 0)$, $B = (x, 0)$, $C = (0, -y)$, and $D = (0, y)$ for integers x and y with $1 \leq x, y \leq 10$ and $2 \mid x + y$. Let T denote the set of all such ellipses with $2 \nmid x + y$. The positive difference between the sum of the areas of all ellipses in T and the sum of the areas of all the ellipses in S can be expressed as $m\pi$, where m is a positive integer. Find m.

Answer: $\boxed{25}$

Solution: Note that the ellipse with axes AB and CD has semi-axes of lengths x and y, so its area is πxy. So, the sum of the areas of the ellipses in S minus the sum of the areas

of the ellipses in T is

$$\sum_{x=1}^{10}\sum_{y=1}^{10}(-1)^{x+y}\pi xy = \left(\sum_{i=1}^{10}(-1)^i i\right)^2 \pi = 5^2\pi,$$

meaning $m = \boxed{25}$.

SR 8: Bob has two coins; one is fair, and one lands on heads with a probability of $\frac{2}{3}$. Bob chooses one of the coins uniformly at random and flips it twice. Alice watches the two coin flips and guesses whether Bob flipped the fair or rigged coin. Given that Alice is a good mathematician and guesses the more likely option (guessing randomly when they are equally likely), the probability she guesses right can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{115}$

Solution: Let $P(f,k)$ and $P(r,k)$ denote the probabilities that the fair coin and the rigged coin, respectively, show k heads. Note that the probability that k heads are shown is $\frac{P(f,k)+P(r,k)}{2}$. If k heads are shown, the probability that Alice guesses correctly is $\frac{\max(P(f,k),P(r,k))}{P(f,k)+P(r,k)}$. So, the answer is

$$\sum_{k=0}^{2}\left(\frac{P(f,k)+P(r,k)}{2}\right)\left(\frac{\max(P(f,k),P(r,k))}{P(f,k)+P(r,k)}\right) = \frac{1}{2}\sum_{k=0}^{2}\max(P(f,k),P(r,k)).$$

It is easy to compute

$$P(f,0)=\frac{1}{4},\quad P(f,1)=\frac{1}{2},\quad \text{and}\quad P(f,2)=\frac{1}{4},$$

and

$$P(r,0)=\frac{1}{9},\quad P(r,1)=\frac{4}{9},\quad \text{and}\quad P(r,2)=\frac{4}{9}.$$

Therefore, the answer is

$$\frac{1}{2}\left(\frac{1}{4}+\frac{1}{2}+\frac{4}{9}\right)=\frac{43}{72} \implies 43+72=\boxed{115}.$$

SR 9: Let a, b, c, and d be positive integers such that $abcd = a+b+c+d$. Find the maximum possible value of a.

Answer: $\boxed{4}$

Solution: Note that $(a,b,c,d) = (4,2,1,1)$ satisfies the equation. Now, suppose that a solution has $a \geq 5$, and let $g(b,c,d) = abcd - a - b - c - d$. Since

$$g(b+1,c,d) - g(b,c,d) = acd - 1 > 0,$$

g is increasing in b, and by symmetry, it is also increasing in c and d. If $b=c=d=1$, then $g(1,1,1) = -3 \neq 0$. Otherwise, WLOG $b \geq 2$, so

$$g(b,c,d) \geq g(2,1,1) = a - 4 \geq 1,$$

a contradiction. In conclusion, the maximum possible value of a is $\boxed{4}$.

SR 10: Let $a_1, a_2, \ldots, a_{14}$ be the roots of $(x^7 - x^3 + 2)^2 = 0$, counted with multiplicity. Find the value of $\prod_{i=1}^{14}(a_i^7 + 1)$.

Answer: $\boxed{4}$

Solution: Let $r_1, r_2, \ldots, r_7$ be the roots of

$$f(x) = x^7 - x^3 + 2 = \prod_{i=1}^{7}(x - r_i).$$

Since each r_i appears exactly twice among $a_1, a_2, \ldots, a_{14}$, we have

$$\prod_{i=1}^{14}(a_i^7 + 1) = \left(\prod_{i=1}^{7}(r_i^7 + 1)\right)^2.$$

Now, we have $r_i^7 - r_i^3 + 2 = 0 \implies r_i^7 + 1 = r_i^3 - 1$. Let ω be a primitive cube root of unity, so that $\omega^3 = 1$, $\omega \neq 1$, and the roots of $x^3 - 1$ are $1, \omega, \frac{1}{\omega}$. So,

$$\begin{aligned}
\prod_{i=1}^{7}(r_i^7 + 1) &= \prod_{i=1}^{7}(r_i^3 - 1) \\
&= \prod_{i=1}^{7}(r_i - 1)\prod_{i=1}^{7}(r_i - \omega)\prod_{i=1}^{7}\left(r_i - \frac{1}{\omega}\right) \\
&= -\prod_{i=1}^{7}(1 - r_i)\prod_{i=1}^{7}(\omega - r_i)\prod_{i=1}^{7}\left(\frac{1}{\omega} - r_i\right) \\
&= -f(1)f(\omega)f\left(\frac{1}{\omega}\right) \\
&= -2(\omega^7 - \omega^3 + 2)\left(\left(\frac{1}{\omega}\right)^7 - \left(\frac{1}{\omega}\right)^3 + 2\right) \\
&= -2(\omega - 1 + 2)\left(\frac{1}{\omega} - 1 + 2\right) \\
&= -2(\omega + 1)\left(\frac{1}{\omega} + 1\right) \\
&= -2\left(1 + \left(\omega + \frac{1}{\omega}\right) + 1\right) \\
&= -2\,(1 - 1 + 1) = -2.
\end{aligned}$$

Thus, our answer is $(-2)^2 = \boxed{4}$.